

Trees

What is a Tree?

A tree is a special type of graph. It consists of nodes (or vertices) connected by edges, and it satisfies two key structural properties:

- The graph is connected.
- The graph is acyclic (it contains no cycles).

Because of these properties, there is exactly one simple path between any two vertices in a tree.

![[tree-example.png | 300]]

Rooted Trees

A rooted tree is a tree where one vertex is chosen as the **root**. This gives the tree a hierarchical structure, where relationships are viewed relative to the root.

In other words, a rooted tree is: - A connected and acyclic graph - With a designated root node

Terminology

Given a rooted tree:

- **Parent:** If a node u is directly above node v , then u is the parent of v .
- **Child:** If u is the parent of v , then v is a child of u .
- **Ancestor:** Any node on the path from a node to the root (including parent, grandparent, etc.).
- **Descendant:** Any node in the subtree below a node.
- **Leaf:** A node with no children.
- **Internal node:** A node with at least one child.
- **Path:** A sequence of nodes where consecutive nodes are connected by edges.

These concepts let us reason about tree structure as a hierarchy instead of just a generic graph.

Depth and Height

For a rooted tree, depth and height are defined using path lengths (number of edges):

- **Depth of a node v :** length of the path from v to the root.
- **Height of a node v :** length of the longest path from v down to a descendant leaf.

For the whole tree T :

- **Depth of T = Height of T** = length of the longest path from the root to a leaf.

Binary Tree

A binary tree is a rooted tree with an additional constraint:

- Each node has at most two children.
- These children are called the **left child** and **right child**.

Binary trees are one of the most important tree classes in algorithms and data structures.

![[binary-tree-example.png | 400]] ## Complete and Almost Complete Binary Trees ### Complete Binary Tree A complete binary tree is a binary tree where all levels are full.

Almost Complete Binary Tree

An almost complete binary tree is a complete binary tree with 0 or more right-most leaves deleted.

Related

- [[Graphs]]
- [[Minimum Spanning Tree (MST)]]